

Inhaled bronchodilators for cystic fibrosis (Review)

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[Intervention Review]

Inhaled bronchodilators for cystic fibrosis

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ABSTRACT

Background

Recurrent wheeze and breathlessness are common in people with cystic fibrosis, and bronchodilators are commonly prescribed. Despite their wide-scale and often long-term use, there is limited objective evidence about their efficacy in cystic fibrosis (CF).

Objectives

To evaluate the effectiveness of inhaled bronchodilators in people with CF.

Search methods

We searched the Cochrane Cystic Fibrosis and Genetic Disorders Group Trials Register comprising references identified from comprehensive electronic databases searches, and handsearches of relevant journals and abstract books of conference proceedings.

Latest search of the Group's Trials Register: 09 September 2010.

Selection criteria

Randomised or quasi-randomised trials comparing inhaled bronchodilators to placebo or another inhaled bronchodilator in people with CF, diagnosed clinically and by sweat or genetic testing and at all stages and severity of lung disease.

Data collection and analysis

The authors independently extracted data and assessed trial quality. A trial's primary author was contacted for any missing data where possible. Data were grouped into classes of bronchodilator and, for each class, into short-term effects (less than one week) and long-term effects (greater to or equal to one week).

Main results

The search identified 66 references; eighteen trials (total of 369 participants) were suitable for inclusion. All but two of the trials were cross-over in design; a meta-analysis was not possible. There were varied conclusions from the different trials reflecting their heterogeneity. Compared to placebo, short-acting beta-2 agonists increased forced expiratory volume at one second (FEV₁) in the short term in three out of five trials, and in the long-term increased peak expiratory flow rate in individuals who had been shown to have bronchial hyperreactivity, bronchodilator responsiveness or both. Compared to placebo, long-acting beta-2 agonists increased FEV₁ and forced expiratory flow between 25% and 75% of expiratory flow in the short term in participants known to have bronchodilator responsiveness, but produced inconsistent results in long-term trials. Short acting-anticholinergics had no consistent effect on lung function tests in either the short or the long term. We found no published trials of fenoterol, formoterol or tiotropium.

Authors' conclusions

It was not possible to determine fully the effectiveness of inhaled bronchodilators in CF as a meta-analysis was not possible. However, short and long-acting beta-2 agonists can be beneficial both in the short and long term in individuals with demonstrable bronchodilator responsiveness or bronchial hyper-responsiveness. As we found no evidence, the use of fenoterol, formoterol or tiotropium in CF cannot be supported.

PLAIN LANGUAGE SUMMARY

Inhaled drugs to open up the airways of people with cystic fibrosis

People with cystic fibrosis are often given drugs to open up their airways when they are breathless or wheezy. The drugs are given either by metered dose inhalers or nebulisers. We looked at how well these drugs worked in both children and adults. We included 18 trials in the review. Some of these used beta-2 agonist bronchodilators and some used anticholinergic bronchodilators. Data for both long-acting and short-acting versions of both drug classes were collected. The results were split into short-term effects (less than one week) and long-term effects (one week or longer). The studies were too varied to allow results to be combined. Both short-acting and long-acting beta-2 agonists improved lung function in the short-term, but only in those people whose airways were found to be sensitive to the bronchodilator after their first dose. Results were not consistent in the long-term. It is reasonable to suggest that a long-term trial of inhaled bronchodilators be considered before their long-term use is recommended. Further large trials are needed to make clear the benefits of these drugs.

BACKGROUND

Description of the condition

Cystic fibrosis (CF) is the commonest autosomal recessive condition in Caucasians, affecting 1 in 2500 births ([CF Foundation 2000](#)). Respiratory disease remains the major cause of morbidity and mortality in CF. Many different therapies are used in the management of respiratory problems: these include oral, intravenous and nebulised antibiotics; physiotherapy; dornase alfa; inhaled corticosteroids; and bronchodilators.

Recurrent wheeze and other features of airway obstruction are common in both children and adults with CF, and bronchodilators are among the most widely prescribed medications for these people. In a large survey in the United States, more than 82% of the participants studied used inhaled bronchodilators on a daily basis ([Brand 2000](#); [Konstan 1999](#)). Several different mechanisms can contribute to wheeze and reversible airways obstruction in CF, including airways mucosal oedema caused by infections and inflammation; stimulation of autonomic nerve fibres as the result of epithelial damage; contraction of airway smooth muscle, caused by mediators of airway infection; inflammation and autonomic nerve stimulation; bronchiectasis; and airway collapse ([Cropp 1996](#)).

The diagnosis of asthma is a very difficult as it is a clinical one and there is no gold standard, but it is based on the presence on symptoms (wheeze, shortness of breath, chest tightness, cough)

and reversible airflow obstruction or bronchial hyper-responsiveness on lung function testing ([BTS 2008](#)). Diagnosis is also different in adults and children, although there are many shared features. Inhaled bronchodilators benefit individuals' symptoms and, in the short term, at least partially reverse airflow obstruction in asthma ([BTS 2008](#)). In CF the mechanisms of airway inflammation differ from asthma and result in progressive non-reversible airflow obstruction. However, a proportion of people with CF will have evidence of asthma on lung function testing (CF asthma) and are atopic, particularly with sensitisation to aspergillus and other moulds. The diagnosis and definition of CF asthma is difficult. A history of wheezing is not specific but can be suggestive if it occurs at night or with exercise. Spirometry is not very helpful in differentiating between the two diseases. Bronchodilator responsiveness, classically used to define asthma, can be misleading, as individuals with CF have an inherent variation in their lung function response. The 'North American Epidemiologic Study of Cystic Fibrosis' (ESCF) suggested that CF asthma is present when there are "episodes of acute airway obstruction reversed by bronchodilators, a strong family history of asthma and/or evidence of atopy, or laboratory evidence of allergy such as eosinophilia or elevated IgE". They reported a prevalence of 31.5% ([Balfour-Lynn 2002](#)). Although CF and atopy, with or without asthma, may co-exist, it is not known if these individuals will derive a greater benefit from the use of inhaled bronchodilators.

Description of the intervention

There are two distinct classes of inhaled bronchodilators, the beta-2 agonists and the anticholinergics. Each of these have short-acting and long-acting agents: short-acting beta-2 agonists (salbutamol or albuterol, terbutaline, fenoterol), long-acting beta-2 agonists (salmeterol, formoterol), short-acting anticholinergic (ipratropium bromide), and the long-acting anticholinergic (tiotropium). It is not appropriate to group these different agents together as they have different properties and mechanisms of action.

It is possible to deliver both these classes of bronchodilator into the lungs via two methods. Firstly, using a dry powder device or a metered dose inhaler with or without the use of a large volume spacer to increase deposition in the lungs; and secondly, as an aerosol via a nebuliser which usually requires a larger starting dose but delivers an equivocal dose to the lungs. This second method has the added advantage of being able to supply oxygen, if needed, at the same time.

For the purpose of this review 'inhaled' shall imply any route that delivers the bronchodilator to the lungs via inhalation.

How the intervention might work

As well as causing relaxation of the bronchial wall muscle, inhaled bronchodilators have effects on mucociliary clearance (Brand 2000) and bacterial damage to the bronchial mucosa (Dowling 1998; Dowling 1999). Bronchodilators are widely prescribed in CF for a variety of other reasons. These include the relief of symptoms directly attributable to CF asthma, to be an adjunct to physiotherapy, to help improve sputum clearance and to prevent symptoms and bronchoconstriction during nebulised therapies, e.g. tobramycin. Their effectiveness in these situations has not been reviewed. All bronchodilators have the potential to worsen rather than improve airway function. This 'paradoxical bronchoconstriction' is the result of the abnormal airway compliance associated with the bronchiectasis in severe lung disease in CF.

Why it is important to do this review

Despite the wide-scale and often long-term use of bronchodilators in CF, there is little objective evidence about their efficacy. There are studies that have assessed the physiological response to bronchodilators over a matter of hours, but the implications of these studies for clinical practice are often unclear (Brand 2000).

OBJECTIVES

This review aims to:

1. evaluate the effectiveness of inhaled bronchodilators in children and adults with CF;
2. assess the most effective drug and dose regimen, if bronchodilators are shown to be effective, and to identify the features that predict a response.

METHODS

Criteria for considering studies for this review

Types of studies

Randomised (RCT) or quasi-randomised trials.

Types of participants

Children, adolescents and adults with CF, diagnosed clinically and by sweat or genetic testing. People with all stages and severity of lung disease.

Types of interventions

Inhaled bronchodilators, delivered by any device, at any dose, at any frequency and for any duration compared to either placebo or another inhaled/nebulised bronchodilator in people with CF.

The bronchodilators included were:

1. short-acting beta-2 agonists (salbutamol/albuterol, terbutaline and fenoterol)
2. long-acting beta-2 agonists (salmeterol and formoterol)
3. short-acting anticholinergic agents (ipratropium bromide)
4. long-acting anticholinergics agents (tiotropium) (*Post hoc* change added after the protocol as the therapy has become licensed for use)

For this review 'inhaled' included the use of pressurised metered dose inhalers (MDIs) with or without a large volume spacer, dry powder devices and nebulisers.

Types of outcome measures

Primary outcomes

1. Change in lung function from baseline
 - i) forced expiratory volume in one second (FEV₁) (litres and per cent predicted)
 - ii) forced vital capacity (FVC) (litres and percent predicted)
 - iii) forced expiratory flow between 25% and 75% of expiratory flow (FEF₂₅₋₇₅) (litres per second and per cent predicted)

- iv) peak expiratory flow rate (PEFR) (litres per minute and per cent predicted)
- v) measures of bronchial hyper-responsiveness and bronchodilator responsiveness
- vi) any other measurements of lung function that are relevant

This assessment included details of the proportion of individuals in whom inhaled bronchodilators produced a deterioration of lung function, as well as those in whom there was either an improvement or no change.

Secondary outcomes

1. Change in the rates of respiratory symptoms
2. Change in quality of life measures (QoL) using standardised and validated QoL scores
3. Change in rates of exacerbations of respiratory symptoms
4. Change in mortality rates
5. Other adverse effects (especially those associated with inhaled bronchodilators i.e. tremor, increased heart rate, dry mouth, increased wheezy and shortness of breath)

Search methods for identification of studies

Electronic searches

Relevant trials were identified from the Group's Cystic Fibrosis Trials Register using the terms bronchodilator AND inhaled. The Cystic Fibrosis Trials Register has been compiled from electronic searches of the Cochrane Central Register of Controlled Trials (Clinical Trials) (updated each new issue of *The Cochrane Library*), quarterly searches of MEDLINE, a search of EMBASE to 1995 and the prospective hand searching of two journals - *Pediatric Pulmonology* and the *Journal of Cystic Fibrosis*. Unpublished work has been identified by searching the abstract books of three major cystic fibrosis conferences: the International Cystic Fibrosis Conference; the European Cystic Fibrosis Conference and the North American Cystic Fibrosis Conference. For full details of all searching activities for the register, please see the relevant sections of the [Cystic Fibrosis and Genetic Disorders Group Module](#). Date of the most recent search of the Group's Cystic Fibrosis Trials Register: 09 September 2010.

Searching other resources

In addition the authors wrote to the major pharmaceutical companies requesting relevant 'data on file'. Where data were missing from selected trials, the authors were contacted directly for more information.

Data collection and analysis

Selection of studies

The three authors (Drs Halfhide, Evans and Couriel) independently selected the references to be included in the review from those identified by the search. No further references were identified by contacting the pharmaceutical companies.

Data extraction and management

Drs Halfhide and Evans independently extracted the data using a specially designed data extraction form approved by the Cochrane Cystic Fibrosis and Genetic Disorders Review Group. Only once was there a difference of opinion regarding the suitability of a reference for inclusion in the review or on its quality; at this point Dr Couriel was asked for a deciding opinion. If information was unclear or incomplete, the primary investigator of the study was contacted where possible.

For this version of the review, we initially reported all the classes of bronchodilator against a placebo and then against the other classes of bronchodilator in turn, e.g. short-acting beta-2 agonist versus placebo, short-acting beta-2 agonist versus long-acting beta-2 agonists. This gave 10 separate classes of results that were reported. For each class we reported the primary and secondary outcomes in two subgroups either as a short-term effect (less than one week) or a long-term effect (equal to or greater than one week). We further divided each of the groups into single versus multiple doses of the agent being tested.

Assessment of risk of bias in included studies

In order to establish a risk of bias for the included studies, Drs Halfhide and Evans assessed the methodological quality of each reference based on a method used by Schultz ([Schulz 1995](#)). In particular, they examined the references for: details of randomisation; whether there had been allocation concealment; whether the trial was blinded for receiver, provider and analyser; whether an intention-to-treat analysis was performed or an intention-to-treat principle was used; and if the number of participants lost to follow up or subsequently excluded from the trial was recorded. We categorised the generation of the randomisation sequence and the allocation concealment as adequate, unclear or inadequate. This translates to a low risk of bias for the 'adequate' category, an unclear risk of bias for the 'unclear' category or a high risk of bias for the 'inadequate' category.

The generation of the randomisation sequence was deemed adequate if the eligibility of the potential participant was unknown to the recruiting person until the decision about assignment had been revealed, ideally by a person not responsible for recruiting participants, for example by using one of the following methods:

a central code or table; on-site computer with allocations not revealed until participants had been enrolled or sequentially numbered; and sealed opaque envelopes. If no details of randomisation were given, it was deemed unclear. Any other methods were deemed inadequate.

Allocation concealment was deemed adequate if the randomisation process was administered by someone who was not responsible for the recruitment of participants. If no methods of recruitment were mentioned, then allocation concealment were deemed unclear. If treatments were not allocated it was deemed inadequate and thought to carry a potential risk of bias.

The blinding of the receiver, provider and analysers were categorised as adequate, unclear or inadequate. As above, this translates to a low risk of bias for the 'adequate' category, an unclear risk of bias for the 'unclear' category or a high risk of bias for the 'inadequate' category. Blinding of receiver, provider and analysers was deemed adequate if a method was stated which prevented the receiver, provider or analysers knowing which intervention arm the participants were in.

Intention-to-treat analysis was assessed using the following: all participants entering the trial; 15% or fewer excluded; more than 15% excluded; or not analysed as intention-to-treat.

Measures of treatment effect

We have not yet been able to enter data in a meta-analysis, but for future updates of this review, if more data are available for inclusion in a meta-analysis, we intend to carry out the following. For the binary outcome measures, we intend to calculate a pooled estimate of the treatment effects for each outcome across trials (the odds of an outcome among treatment allocated participants to the corresponding odds among controls) using risk ratio.

For the continuous data, we plan to record the mean change from baseline for each group or the mean post-treatment or post-intervention values and standard deviation for each group. We intend to calculate a pooled estimate of treatment effect using the mean difference and 95% confidence intervals.

Unit of analysis issues

Of the included trials, two were parallel trials (Sheehan 2003; Turchetta 1998); the other 16 used a cross-over design. Ideally we would have liked to use the inverse variance methods that are recommended by Elbourne when conducting a meta-analysis combining results from cross-over trials (Elbourne 2002). However, due to restrictions on the data that were available from the papers, the only method that we would have been able to use would be to treat the cross-over trials as if they were parallel trials. Elbourne says that this approach will produce conservative results as it does not take into account within-patient correlation (Elbourne 2002). Also each participant will appear in both the treatment and control group, so the two groups will not be independent. Since we

believe the data will have been analysed more appropriately by the primary authors, we have reported the results that were obtained from the primary paper and not entered the data from the cross-over trials into the [Data and analyses](#) section (which would have enabled us to calculate a pooled effect of treatment).

Dealing with missing data

We contacted the primary authors of the studies for any missing data.

For future updates of this review, in order to allow an intention-to-treat analysis, we will seek data on the number of participants by allocated treatment group, irrespective of compliance and whether or not the participant was later thought to be eligible or otherwise excluded from treatment or follow up.

Assessment of heterogeneity

When sufficient trials that can be used in a meta-analysis are included in the review (at least four), we plan to test for heterogeneity using a standard chi-squared test and the I^2 statistic (Higgins 2003). The I^2 statistic describes the percentage of total variation across trials which is due to heterogeneity rather than chance (Higgins 2003). Its values range from 0% to 100%, and a simplified categorization of heterogeneity that we plan to use is of low (I^2 value of 25%), moderate (I^2 value of 50%), and high (I^2 value of 75%) (Higgins 2003).

Assessment of reporting biases

When sufficient trials are included in the review, we will attempt to identify any publication bias in the included trials by generating a funnel plot. We will also attempt to identify any selective reporting in the included publications, ideally by comparing the trial protocols with the final papers, but also by careful examination of the trial publications and consideration of reporting of both positive and negative effects of the intervention. We will look at the sponsors and competing interests of the authors to determine the role of external bias being introduced.

Data synthesis

When sufficient trials are included in the review to produce a meta-analysis and if we identify little or no heterogeneity between trials, we plan to analyse the data using a fixed-effect model. However, if we identify a moderate to high degree of heterogeneity, we will employ a random-effects model in our analysis.

Subgroup analysis and investigation of heterogeneity

If we find significant heterogeneity (over 50%), we will investigate the possible causes of this further by performing subgroup analyses looking at the effect of age, severity of lung disease, the presence

of atopy (defined by allergy skin prick testing or RAST testing) and the presence of reversible airway obstruction (as defined in American Thoracic Society guidelines of lung function testing (ATS 2005)).

Sensitivity analysis

If we are able to include sufficient trials, we will perform a sensitivity analysis on the methodological quality of the trials including and excluding quasi-randomised controlled trials.

RESULTS

Description of studies

See: [Characteristics of included studies](#); [Characteristics of excluded studies](#); [Characteristics of studies awaiting classification](#); [Characteristics of ongoing studies](#).

Results of the search

The search by the Cochrane Cystic Fibrosis and Genetic Disorders Group identified 66 references to 46 trials. The authors were unable to identify any further trials or data, either by further searches of journals or by contacting the major pharmaceutical companies. Only the main significant results will be presented.

There are a total of 26 references to 18 included trials (369 participants - both children and adults with CF) (Avital 1992; Dodd 2005; Duncan 1982; Eggleston 1991; Hordvik 1996; Hordvik 1999; Hordvik 2002; Konig 1998; Pitcher 1982; Salvatore 2002; Sanchez 1992; Sanchez 1993; Serisier 2007; Sheehan 2003; Turchetta 1996; Turchetta 1998; Wiebicke 1990; Ziebach 2001). It was not possible to extract data from the Duncan trial, despite trying to contact the primary authors (Duncan 1982).

There are two references to one trial which has been listed as 'Awaiting classification' (Bessaci 2008). Currently there are two abstracts published which refer to this study, but neither of them are clear as to the randomisation process used in the study. We plan to assess this study again once further details have been published. A further trial is listed as 'Awaiting classification' (NCT00737100). This phase II trial of two doses of tiotropium bromide has been completed, but as yet no data have been published.

There is also one reference to an ongoing placebo-controlled phase III trial of tiotropium bromide which is due to be completed in December 2011 (NCT01179347).

A total of 36 references to 25 trials are listed as excluded (Avital 1991; Chua 1989; Chua 1993; Finnegan 1989; Gulliver 2003; Kattan 1980; Kohler 2004; Konig 1995; Konig 2005; Kovaleva 2000; Kraemer 1996; Larsen 1979; Liedtke 1996; Maayan 1989; Marks 1998; Mortensen 1993; Nikolaizik 1996; Orlik 2002;

Peckham 1995; Robuschi 1990; Shapiro 1976; Sivan 1990; Tecklin 1976; Van Asperen 1988; Weintraub 1989).

Included studies

Type of bronchodilator

A variety of bronchodilators were used in the included trials. Full details of the dose, duration, delivering device and frequency are given in the [Characteristics of included studies](#). A summary of this information is presented in the additional tables ([Table 1](#)).

1. Short-acting beta-2 agonists

Albuterol was used in seven trials (Eggleston 1991; Hordvik 1996; Hordvik 1999; Hordvik 2002; Konig 1998; Sanchez 1992; Sheehan 2003), and salbutamol in eight trials (Avital 1992; Dodd 2005; Duncan 1982; Pitcher 1982; Turchetta 1996; Sanchez 1993; Serisier 2007; Ziebach 2001). Albuterol and salbutamol are the same biochemical molecule but salbutamol is the generic name used in the UK and albuterol is used in the USA. For this review the name of the drug given by the primary author will be used.

The dose of nebulised salbutamol (or albuterol) ranged from 2.5 mg (Avital 1992; Hordvik 1996; Hordvik 2002) to 5.0 mg (Duncan 1982; Sanchez 1992) and was age-specific in two trials (Pitcher 1982; Ziebach 2001). The dose of salbutamol via a metered dose inhaler (MDI) ranged from 200 µg (Sanchez 1993) to 600 µg (Serisier 2007). The dose of albuterol via an MDI was 180 µg in four trials (Eggleston 1991; Hordvik 1999; Konig 1998; Sheehan 2003). Two trials did not mention either the dose or the device used in the trial (Dodd 2005; Turchetta 1996).

The frequency of treatment was once daily (Avital 1992; Dodd 2005; Duncan 1982; Pitcher 1982; Sanchez 1992; Serisier 2007; Ziebach 2001), twice daily (Hordvik 2002; Konig 1998; Sanchez 1993; Sheehan 2003), three times daily (Hordvik 1996; Turchetta 1996) or four times daily. (Eggleston 1991; Hordvik 1999).

There were no trials using terbutaline or fenoterol.

2. Long-acting beta-2 agonist

Salmeterol was used in five trials (Hordvik 1999; Hordvik 2002; Salvatore 2002; Turchetta 1996; Turchetta 1998). The dose of salmeterol via an MDI varied from 42 µg (Hordvik 1999) to 84 µg (Salvatore 2002) up to 100 µg (Hordvik 2002). The dose was unspecified in two trials (Turchetta 1996; Turchetta 1998). The frequency was once daily at night in one trial (Salvatore 2002) and twice daily in four trials (Hordvik 1999; Hordvik 2002; Turchetta 1996; Turchetta 1998).

There were no trials using formoterol.

3. Short-acting anticholinergics agents

Ipratropium was used in six trials (Avital 1992; Pitcher 1982; Sanchez 1992; Sanchez 1993; Wiebicke 1990; Ziebach 2001). The dose of nebulised ipratropium was 0.25 mg in three trials (Avital 1992; Sanchez 1992; Wiebicke 1990) and age-specific in two trials (Pitcher 1982; Ziebach 2001). The dose of inhaled ipratropium via an MDI was 80 µg in another trial (Sanchez 1993). The frequency was once daily in all six trials (Avital 1992; Pitcher 1982; Sanchez 1992; Sanchez 1993; Wiebicke 1990; Ziebach 2001).

4. Long-acting anticholinergics

No trials used this class of bronchodilator.

Mode of delivery

A variety of modes of delivery were used. Nebulised solutions were compared in eight trials (Avital 1992; Duncan 1982; Hordvik 1996; Hordvik 2002; Pitcher 1982; Sanchez 1992; Wiebicke 1990; Ziebach 2001). An MDI was used in six trials (Eggleston 1991; Hordvik 1999; Konig 1998; Salvatore 2002; Turchetta 1996; Turchetta 1998); three trials additionally used a MDI and a large volume spacer (Sanchez 1993; Serisier 2007; Sheehan 2003). There is no information regarding the device used by Dodd (Dodd 2005).

Trial duration

Short-term effects (defined as less than one week) were reported in thirteen trials (Avital 1992; Dodd 2005; Duncan 1982; Hordvik 1996; Hordvik 1999; Pitcher 1982; Sanchez 1992; Sanchez 1993; Salvatore 2002; Serisier 2007; Sheehan 2003; Wiebicke 1990; Ziebach 2001). Long-term effects (greater or equal to one week) were reported in five trials (Eggleston 1991; Hordvik 2002; Konig 1998; Turchetta 1996; Turchetta 1998).

Participant characteristics

Five trials recruited only children (under 16 years old) (Avital 1992; Pitcher 1982; Sheehan 2003; Turchetta 1996; Turchetta 1998). Eight trials recruited a combination of children and adults (Eggleston 1991; Hordvik 1996; Hordvik 1999; Hordvik 2002; Konig 1998; Sanchez 1992; Sanchez 1993; Ziebach 2001). Three trials recruited adults only (Dodd 2005; Salvatore 2002; Serisier 2007). The age of the participants was unclear in one trial (Duncan 1982).

Thirteen trials commented on disease severity (Avital 1992; Eggleston 1991; Hordvik 1996; Hordvik 1999; Hordvik 2002; Konig 1998; Salvatore 2002; Sanchez 1992; Sanchez 1993; Serisier 2007; Turchetta 1998; Wiebicke 1990). The methods used to assess disease severity varied between trials. Six trials employed the Shwachman score (Avital 1992; Eggleston 1991; Konig 1998;

Sanchez 1992; Sanchez 1993; Wiebicke 1990); five trials used baseline FEV₁ (Hordvik 2002; Salvatore 2002; Serisier 2007; Turchetta 1998; Ziebach 2001); and two trials used the National Institute of Health (NIH) score (Hordvik 1996; Hordvik 1999). The remaining five trials made no comment on the severity of disease in participants (Duncan 1982; Dodd 2005; Pitcher 1982; Sheehan 2003; Turchetta 1996). No participants were excluded as a result of disease severity.

Only one trial tested for the presence of atopy by RAST and in this trial, eight of the twenty-two participants had more than one positive skin test (Eggleston 1991). A further trial mentions that one of the eleven children randomised had a positive family history of hay fever (Avital 1992). No other trial mentions family history or personal history of atopy.

Six trials commented on reversible airway disease, either from participant history or testing for it (Avital 1992; Eggleston 1991; Sanchez 1992; Sanchez 1993; Salvatore 2002; Serisier 2007). In one trial, all randomised children had a history of wheezing (Avital 1992). Another trial subdivided 27 participants, who responded to a methacholine challenge (16 responders) from those who did not (11 non-responders) (Eggleston 1991). In the first Sanchez trial, 9 of the 12 randomised participants experienced an increase in FEV₁ greater than 10% in response to a beta-2 agonist on more than two occasions (Sanchez 1992). In the second Sanchez trial all randomised participants had an increase of more than six per cent in FEV₁ in response to a bronchodilator (Sanchez 1993). A further trial subdivided the 23 randomised participants into 14 responders if they had an increase greater than 12% in FEV₁ after albuterol (Salvatore 2002). In the final trial, all randomised participants had a histamine challenge to determine responders, but no data were available (Serisier 2007). Three trials used a pre-defined improvement in FEV₁ to determine participants in two trials (Sanchez 1992; Sanchez 1993); and a subclass called responders in a further trial (Salvatore 2002). This improvement in FEV₁ was 6% in one trial (Sanchez 1993), 10% in a second trial (Sanchez 1992) and 12% in a third trial (Salvatore 2002). The general recommendation is that a significant response is between 10% and 15%; and a level of below this may be accounted for by an error of measurement (Sanchez 1993).

Excluded studies

A total of 25 trials were excluded for a range of reasons. Eight trials did not report on our pre-defined outcomes (Chua 1989; Chua 1993; Kohler 2004; Konig 2005; Liedtke 1996; Mortensen 1993; Orlik 2002; Peckham 1995). A further nine trials did not employ a pre-defined bronchodilator (Avital 1991; Gulliver 2003; Kattan 1980; Larsen 1979; Maayan 1989; Shapiro 1976; Sivan 1990; Tecklin 1976 Weintraub 1989). Two trials were neither randomised or quasi-randomised controlled trials (Konig 1995; Van Asperen 1988). The remaining five trials were excluded because a route of administration was used which we had not

planned to include (Finnegan 1989); or the trial compared the type of delivery device rather than the drug (Marks 1998); or the intervention drug was not a bronchodilator (Nikolaizik 1996; Robuschi 1990); or, the study examined the relationship of the intervention to physiotherapy (Kraemer 1996). One trial looked at a combination of enzymotherapy and nebulizer therapy using bronchodilators and so was not eligible (Kovaleva 2000).

Risk of bias in included studies

In order to establish the risk of bias, we assessed methodological quality based on a method used by Schulz (Schulz 1995) and related this to the risk of bias in the trials.

Allocation

We judged randomisation to be adequate in 10 trials where the randomisation method used was a central code or table (Avital 1992; Eggleston 1991; Hordvik 1996; Hordvik 1999; Hordvik 2002; Konig 1998; Sanchez 1992; Sanchez 1993; Serisier 2007; Ziebach 2001). Randomisation was unclear in eight trials (Dodd 2005; Duncan 1982; Pitcher 1982; Salvatore 2002; Serisier 2007; Sheehan 2003; Turchetta 1996; Turchetta 1998).

The allocation concealment was rated 'adequate' in 11 trials since the intervention was not revealed until after the participants had been recruited (Avital 1992; Dodd 2005; Eggleston 1991; Hordvik 1996; Hordvik 1999; Hordvik 2002; Konig 1998; Sanchez 1992; Sanchez 1993; Serisier 2007; Ziebach 2001). Allocation concealment was deemed unclear in the other seven trials (Duncan 1982; Pitcher 1982; Salvatore 2002; Sheehan 2003; Turchetta 1996; Turchetta 1998; Wiebicke 1990).

Blinding

Blinding for receiver, provider and analysers was adequate in ten trials (Avital 1992; Eggleston 1991; Hordvik 1999; Hordvik 2002; Konig 1998; Sanchez 1992; Sanchez 1993; Salvatore 2002; Serisier 2007; Ziebach 2001). The method of blinding in the remaining eight trials was unclear (Dodd 2005; Duncan 1982; Hordvik 1996; Pitcher 1982; Sheehan 2003; Turchetta 1996; Turchetta 1998; Wiebicke 1990). Three of these trials state that the trials were double-blind but no further details given (Dodd 2005; Hordvik 1996; Wiebicke 1990), whilst three trials were stated to be single-blinded (Duncan 1982; Pitcher 1982; Turchetta 1998); blinding was not discussed in two trials (Sheehan 2003; Turchetta 1996). It is unlikely that the Sheehan trial was blinded as the maximum number of puffs from albuterol and placebo was different (Sheehan 2003).

Incomplete outcome data

None of the included trials used an intention-to-treat analysis or an intention-to-treat principle. All participants completed the trial in nine trials (Avital 1992; Duncan 1982; Pitcher 1982; Salvatore 2002; Sanchez 1992; Sanchez 1993; Sheehan 2003; Turchetta 1998; Wiebicke 1990). In four trials there were less than 15% of participants lost to follow up (Hordvik 1996; Hordvik 1999; Serisier 2007; Ziebach 2001). Of these trials, none of the participants dropped out due to side effects. In the earlier Hordvik trial, two of the 24 randomised participants, dropped out as they developed a pulmonary exacerbation needing additional treatment (Hordvik 1996). In the second Hordvik trial, two of the 20 randomised participants dropped out; one did not finish the first trial day, the second was also taking inhaled steroids at the time of enrolment (Hordvik 1999). In the Serisier trial two of the 22 participants who enrolled did not finish the trial (Serisier 2007). One participant declined to participate following enrolment and the second did not complete the second exercise test for personal reasons. In the fourth trial, one of the 17 randomised participants was lost to follow up due to being acutely unwell (Ziebach 2001). We judged all these trials to have a low risk of bias from missing data.

In five trials there were more than 15% of participants lost to follow up (Dodd 2005; Eggleston 1991; Hordvik 2002; Konig 1998; Turchetta 1996). In the Dodd trial, two of the 10 participants who were enrolled were excluded from the analysis; one due to sub-maximal effort during the exercise test and the second for inter-current infection (Dodd 2005). In the Eggleston trial, two of the 27 participants were dropped for non-compliance, and three for pulmonary exacerbations requiring additional treatment (compliance can be defined as the failure of the participant to take their medication as prescribed) (Eggleston 1991). In the Hordvik trial, eight of the 44 randomised participants were lost to follow up: one participant dropped out during the 'run in' period; and seven during the intervention arm (four for non-compliance and three due to increased respiratory problems) (Hordvik 2002). In the Konig trial, there were nine withdrawals from 30 participants: four for poor compliance; three at parent's or participant's request; one for an exacerbation; and one for starting additional treatment (Konig 1998). In the Turchetta trial, four of the 22 enrolled children did not complete the study, but no information is given to the reasons for this (Turchetta 1996). Konig compared pulmonary function tests in those that were lost to follow up (Konig 1998); Dodd, Eggleston, Hordvik and Turchetta do not comment on the lung function tests of those lost to follow up (Dodd 2005; Eggleston 1991; Hordvik 2002; Turchetta 1996). We judged these trials to have a high risk of bias from missing data.

Selective reporting

We were not able to identify any selective reporting in any of the 18 included trials.

Other potential sources of bias

We did not identify any other sources of bias, e.g. funding by the pharmaceutical company, in the included trials.

Effects of interventions

Although 18 trials were eligible for inclusion, it was not possible to extract data for analysis from one of these (Duncan 1982). We present the outcomes for which we have been able to extract information from the included studies. We were unable to enter data into a meta-analysis as there was wide heterogeneity in the outcomes reported and the methodology used (for further details see the section [Description of studies](#) and the table [Characteristics of included studies](#)).

SHORT-ACTING BETA-2 AGONISTS

Short acting beta-2 agonist compared to placebo

Primary outcome

1. Changes in lung function from baseline

a. FEV₁

i. Short term

In the short term, seven trials comparing salbutamol (albuterol) to placebo presented data for FEV₁ (Avital 1992; Dodd 2005; Hordvik 1996; Hordvik 1999; Pitcher 1982; Serisier 2007; Ziebach 2001) and one presented data for FEV_{0.5} (Sheehan 2003). In the six trials using a single-dose per day, two showed no significant increase in FEV₁ after salbutamol (Avital 1992; Pitcher 1982) and one reported no significant difference for FEV_{0.5} between the two groups (Sheehan 2003).

The remaining three trials showed varying improvements in FEV₁ (Dodd 2005; Serisier 2007; Ziebach 2001). Two trials compared the effect in relation to exercise (Dodd 2005; Serisier 2007). Dodd showed that the mean (standard deviation (SD)) change in FEV₁ before exercise was greater with salbutamol 9.8% (4.5%) than with saline 0.2% (5.8%) ($P < 0.001$) (Dodd 2005). Serisier showed an increase in FEV₁ (percentage change (SD) pre- and post-exercise) for salbutamol, 18% (12%) compared to saline, 5% (5%) ($P = 0.01$) (Serisier 2007). A third trial, with no exercise, showed a significant increase in FEV₁ (% change (standard error of the mean (SEM))) with salbutamol of, 7% (0.4%) compared to saline, -1% (0.3%) (Ziebach 2001).

Of the two trials using multiple doses of drug per day, the earlier Hordvik trial showed an increase in FEV₁ (percentage change (SEM) across the day with albuterol, 10.1% (2.8%) compared

to saline, 3.9% (1.9%) ($P = 0.036$) (Hordvik 1996). The later Hordvik trial showed a similar increase (Hordvik 1999).

ii. Long term

In the long term, two trials compared multiple doses of albuterol and saline and found no significant change in FEV₁ over four weeks with a four times a day regimen (Eggleston 1991); or over six months with a twice-a-day regimen (Konig 1998).

b. FVC

i. Short term

In the short term, five trials presented data for FVC (Hordvik 1996; Hordvik 1999; Pitcher 1982; Sheehan 2003; Ziebach 2001).

For the trials using a single dose of drug per day, one trial found no significant change in FVC with salbutamol (Pitcher 1982); and one reported a significant improvement, but no data were given (Ziebach 2001).

For the trials using multiple doses of drug per day, the earlier Hordvik trial showed a significant increase in FVC (percentage change (SEM)) across the day with albuterol, 8.1% (1.2%) compared to saline, 3.5% (1.4%) (Hordvik 1996); but in the later Hordvik trial and in the Sheehan trial, there was no significant change in FVC across the day when albuterol was compared to saline (Hordvik 1999; Sheehan 2003).

ii. Long term

In both long-term trials using multiple doses of drug per day, there was no significant difference in FVC (Eggleston 1991; Konig 1998).

c. FEF₂₅₋₇₅

i. Short term

In the short term, five trials presented data for FEF₂₅₋₇₅ (Hordvik 1996; Hordvik 1999; Pitcher 1982; Sheehan 2003; Ziebach 2001).

Of the trials using single daily doses, two trials reported a significant change in FEF₂₅₋₇₅ with salbutamol compared to saline, but no data were given (Pitcher 1982; Ziebach 2001).

In both Hordvik trials, using multiple doses of drug per day, there was no significant difference in FEF₂₅₋₇₅ (Hordvik 1996; Hordvik 1999). However, in the Sheehan trial there was a significantly greater percentage change in FEF₂₅₋₇₅ in those participants receiving albuterol compared to placebo (mean percentage change (SD) albuterol 13% (17%) versus placebo 3% (7%) ($P < 0.02$) (Sheehan 2003).

ii. Long term

In two trials presenting long-term data and using multiple doses of drug per day, there was no effect on FEF₂₅₋₇₅ (Eggleston 1991; Konig 1998).

d. PEF_R

i. Short term

In the short term, only Pitcher presented data for PEF_R; he stated that there was a significant increase but no analysable data were presented (Pitcher 1982).

ii. Long term

In the long term, one trial showed a significant increase in PEFR in those taking four times a day albuterol compared to saline, but only in those who had a pre-determined bronchial hyperreactivity to methacholine (Eggleston 1991). This group was termed responders in the paper. For the responders the mean change in PEFR (percentage change (SD)) over one month was 12% (32%) for albuterol compared to -0.4% (19%) for placebo ($P = 0.05$). For responders there was also a greater maximum increase (SD) in PEFR over one month with albuterol, 45 L/min (42 L/min) compared to placebo, 22 L/min (17 L/min) ($P < 0.05$). For the non-responders the mean change and the maximum PEFR were not different (Eggleston 1991). Konig showed no significant difference in mean PEFR over six months when twice a day albuterol was compared to placebo (Konig 1998).

e. Measure of bronchial hyper-responsiveness

i. Short term

In the short term, one trial looked at the concentration of methacholine needed to produce a drop of 20% in baseline FEV₁ (PC₂₀) with pre-treatment of salbutamol or placebo (Avital 1992). Salbutamol resulted in significantly greater mean PC₂₀ 1.24 mg/ml compared to placebo 0.41 mg/ml, $P < 0.01$. Another trial compared the difference in percentage change of FEV₁ and maximal exercise capacity in those who had a histamine challenge in a sub-analysis and found there was no significant difference in either parameter in those who tested either histamine positive or negative (Serisier 2007).

ii. Long term

In the long term, one trial used the methacholine challenge test to subdivide participants into responders and non-responders (Eggleston 1991). Responders were those where the concentration of methacholine needed to produce a 20% drop in baseline FEV₁ (PC₂₀) was less than 60 breath units (one breath unit equates to one inhalation of 1.0 mg/ml) and non-responders were defined as those where the PC₂₀ was greater than 90 breath units. Of 27 participants, 16 were responders. In these 16 responders bronchial responsiveness was tested using 200 µg of inhaled isoproterenol. Four responders had a greater than 10% increase in baseline FEV₁ but there was no significant difference in these participants in long-term changes in lung function tests (Eggleston 1991).

Secondary outcomes

1. Change in the rates of respiratory symptoms

Two trials commented on respiratory symptoms (Eggleston 1991; Konig 1998). One trial used a daily symptom diary, where participants scored wheeze, cough and exercise tolerance (Eggleston 1991). There was no significant difference between albuterol and placebo in the symptom scores (Eggleston 1991).

5. Adverse effects

Side effects were mentioned in five trials (Dodd 2005; Eggleston 1991; Hordvik 1996; Hordvik 1999; Pitcher 1982). Dodd reports that there was no significant change in breathing pattern, dyspnoea or leg discomfort between the two groups (salbutamol and placebo) on the day of testing (Dodd 2005). Similarly Hordvik commented that no participants experienced side effects (Hordvik 1999). However, in the other three trials, some side effects are reported. In one trial, five out of 22 randomised participants experienced side effects when albuterol was compared to placebo (Eggleston 1991). In this trial, two participants had mild tremors on albuterol, but none on placebo; two experienced fatigue on albuterol and one of these continued with symptoms on placebo; and one participant had palpitations on placebo (Eggleston 1991). The earlier Hordvik trial stated that all participants had an mean increase in pulse rate of four beats per minute on albuterol compared to saline (Hordvik 1996). The final trial randomised 30 participants and there was a significant fall in FEF₂₅₋₇₅ in two children after salbutamol and one after ipratropium (Pitcher 1982).

Short-acting beta-2 agonists compared to anticholinergics

Primary outcome

1. Change in lung function from baseline

a. FEV₁

i. Short term

In the short term, five trials reported data for this outcome (Avital 1992; Pitcher 1982; Sanchez 1992; Sanchez 1993; Ziebach 2001). Four trials used a single dose of drug per day and one trial used twice-a-day doses of salbutamol compared to once-a-day ipratropium (Sanchez 1993). In two trials no extractable data were presented (Pitcher 1982; Ziebach 2001). In two other trials, there was no significant difference in mean change in FEV₁ between salbutamol and ipratropium (Avital 1992; Sanchez 1993). In the final trial, there was an increase in FEV₁ for combined treatment with albuterol and ipratropium compared to single therapy, but there were no extractable data (Sanchez 1992).

b - d. FVC, FEF₂₅₋₇₅, PEFR

i. Short term

In the short term, one trial found no significant difference between the two bronchodilators for these parameters (Sanchez 1993).

e. Measures of bronchial hyper-responsiveness

i. Short term

In the short term, one trial looked at the response to a methacholine challenge (Avital 1992). Ipratropium produced a greater increase in the concentration of methacholine needed to produce a 20% drop in FEV₁ (PC₂₀) 7.37 mg/ml, compared to salbutamol 1.24 mg/ml ($P < 0.0001$).

f. Other measures of lung function

i. Short term

In the short term, two trials measured airway resistance (Raw) (Sanchez 1992; Sanchez 1993) and one trial looked at oxygen saturation (Sanchez 1993). In the earlier Sanchez trial, the Raw decreased with both single treatments but the decrease was greater with combined salbutamol and ipratropium (mean decrease in Raw 32%) $P < 0.025$ (Sanchez 1992). In the later Sanchez trial, there was no significant difference in Raw and oxygen saturation across the day with salbutamol or ipratropium (Sanchez 1993).

Secondary outcomes

1. Change in the rates of respiratory symptoms,

One trial commented that no participants had additional respiratory symptoms (Sanchez 1993).

5. Other adverse effects

One trial reported that amongst 30 children that there was a significant fall in FEF_{25-75} in two children after salbutamol and one after ipratropium (Pitcher 1982). A further trial stated that there were no side effects (Ziebach 2001).

Short-acting beta-2 agonists compared to long-acting bronchodilators

Primary outcomes

1. Change in lung function from baseline

a. FEV_1

i. Short term

In the short term only Hordvik reported on this outcome and found no significant difference in FEV_1 across the day when four times a day albuterol was compared to twice-daily salmeterol (Hordvik 1999). However, from one morning to the next there was a significant difference in FEV_1 (percentage change) with salmeterol, 7.3% compared to albuterol, -0.7% ($P < 0.005$). There was no reported SD or SEM (Hordvik 1999).

ii. Long term

In the long term, two trials compared short-acting to long-acting beta 2 agonists (Hordvik 2002; Turchetta 1998). The Hordvik trial reported that after four weeks there was no significant difference in mean change in FEV_1 between albuterol and salmeterol (Hordvik 2002). However, there was a difference in mean change

in FEV_1 (percentage change (SD)) between albuterol and salmeterol over 12 and 24 weeks; for 12 weeks mean change in FEV_1 (percentage change (SD)) for albuterol was -6.2% (8.5%) compared to salmeterol 1.8% (7.6%) ($P = 0.013$); for 24 weeks the change was for albuterol -6.5% (9.1%) compared to salmeterol 1.7% (5.9%) ($P = 0.002$) (Hordvik 2002). The Turchetta paper reported greater changes in baseline FEV_1 with salmeterol than with salbutamol, but no data are given (Turchetta 1996).

b. FVC

i. Short term

In the short term, Hordvik, reported no significant difference in FVC between multiple doses of salmeterol and albuterol (Hordvik 1999).

ii. Long term

In the long term, in a further Hordvik trial, there was no significant difference in FVC at either 4, 12 or 24 weeks between twice-daily albuterol and salmeterol (Hordvik 2002). However, Turchetta reported greater changes from baseline FVC in those participants on 30 days of salmeterol compared to salbutamol (Turchetta 1996).

c. FEF_{25-75}

i. Short term

In the short term, Hordvik, using multiple doses of drug per day, reported no significant difference in FEF_{25-75} across the day, but there was a change by the next morning of 17% with salmeterol compared to 2.1% with albuterol ($P < 0.03$). No SDs were reported (Hordvik 1999).

ii. Long term

In the long term, Hordvik reported no significant difference in mean change in FEF_{25-75} for 4 or 24 weeks, but there was a difference in FEF_{25-75} (percentage change (SD)) after 12 weeks for twice-daily doses of salmeterol 2.3% (9.6%) compared to albuterol, -10.8% (15.7%), $P = 0.0017$ (Hordvik 2002). In the Turchetta trial, there was again a greater increase in baseline FEF_{25-75} with the salmeterol treatment than with the salbutamol treatment (Turchetta 1996).

Secondary outcomes

1. Changes in rate of respiratory symptoms

ii. Long term

Hordvik commented on respiratory symptoms (Hordvik 2002). The trial used a patient reported symptom diary for congestion, cough, shortness of breath and quality of sleep (Hordvik 2002). Full data were not available, but there was no significant difference in the symptom scores in the trial period at four weeks and overall at 24 weeks between albuterol compared to salmeterol in the long term. There was a lower mean symptom score for salmeterol than albuterol in the last 12 weeks of each treatment period ($P < 0.001$) (Hordvik 2002).

3. Change in rates of exacerbations of respiratory symptoms

ii. Long term

Hordvik also reported the difference in the use of additional antibiotics in the long term (Hordvik 2002). In both the 4- and 24-week treatment periods there were trends towards fewer antibiotic interventions for salmeterol compared to albuterol but this was only significant over the four-week periods ($P = 0.011$). Over 24 weeks there was no significant difference.

5. Adverse effects

Side effects are mentioned in two trials (Hordvik 1999; Hordvik 2002). Details of side effects for the earlier Hordvik trial are mentioned above (Hordvik 1999). The later Hordvik trial reported side effects in the long term (Hordvik 2002). These included shortness of breath, headache, tremor, nervousness and chest pain but there were no significant differences between the two groups (Hordvik 2002).

LONG-ACTING BETA-2 AGONISTS

Long-acting beta-2 agonist compared to placebo

Primary outcome

1. Change in lung function from baseline

a. FEV₁

i. Short term

In the short term, two trials compared salmeterol to placebo (Hordvik 1999; Salvatore 2002). The Hordvik trial, found no significant difference in FEV₁ across the day but there was a significant change in FEV₁ (percentage change (SEM)) when compared the next morning for salmeterol, 7.3% (2.5%) compared to saline, 1.5% (1.4%) ($P < 0.05$) (Hordvik 1999).

In the other trial, there was a significant change in FEV₁ overnight in those already known to have a greater than 12.5% rise in FEV₁ to albuterol, called responders in the paper (Salvatore 2002). In the responders, the mean (SD) morning FEV₁ was in those on once a day salmeterol, delivered at night, 55.1% (13.8%) compared to those on saline, 49.8% (12.9%) ($P < 0.0001$). This was a significant increase in mean morning FEV₁ with salmeterol of 10.6% compared to placebo. There was no significant difference in the non-responders (Salvatore 2002).

ii Long term

One trial investigated long-term effects and reported that there was a significantly greater increase in FEV₁ from baseline in the participants who received salmeterol, two puffs twice a day, than

those who received placebo, but no data were given (Turchetta 1998).

b. FVC

i. Short term

In the short term, one trial reported data for FVC; and there was no significant difference across the day or until the next morning between salmeterol and placebo (Hordvik 1999).

c. FEF₂₅₋₇₅

i. Short term

In the short term, Hordvik showed no significant improvement across the day or the next day between salmeterol and placebo (Hordvik 1999). However, Salvatore, showed a significant improvement in mean (SD) morning FEF₂₅₋₇₅ % predicted in those on salmeterol who were known responders, 26.7% (10.5%) compared to saline, 21.3% (8.9%) ($P < 0.0001$) (Salvatore 2002).

f. Other measures of lung function

i. Short term

One trial reported oxygen saturation in the short term (Salvatore 2002). In the subgroup termed responders the mean nocturnal oxygen saturation was significantly higher with salmeterol (mean (SD) oxygen saturation 93.6% (2.2%)) than with placebo (92.4% (2.8%)) ($P < 0.0005$). There was no significant difference in the non-responders (Salvatore 2002).

Secondary outcomes

5. Adverse effects

Only one trial stated that there were no side effects experienced (Hordvik 1999).

Long-acting beta-2 agonists compared to short-acting beta agonists

Please see the comparison listed in the section above 'Short-acting beta agonists'.

Long acting beta-2 agonists compared to anticholinergics

No trial looked at this class in either the short or long term.

SHORT-ACTING ANTICHOLINERGICS

Short-acting anticholinergics compared to placebo

Primary outcomes

1. Change in lung function from baseline

a. FEV₁

i. Short term

In the short term, four trials compared once-a-day ipratropium (Avital 1992; Pitcher 1982; Wiebicke 1990; Ziebach 2001). Three trials found no significant difference in the changes in FEV₁ when compared to placebo (Avital 1992; Pitcher 1982; Wiebicke 1990). The third trial found an improvement in FEV₁, percentage change (SEM) for ipratropium, 6% (0.5%) compared to saline, 1% (0.1%), $P = 0.0045$ (Ziebach 2001).

b. FVC

i. Short term

In the short term there was no change in FVC in the two trials that reported data (Pitcher 1982; Ziebach 2001).

c. FEF_{25–75}

i. Short term

In the short term, two trials reported a significant improvement but there were no extractable data (Pitcher 1982; Ziebach 2001); one trial showed no significant change in FEF_{25–75} (Wiebicke 1990).

Secondary outcomes

5. Adverse effects

One trial randomised 30 participants and stated that there was a significant fall in FEF_{25–75} in two children after salbutamol and one after ipratropium (Pitcher 1982).

Short-acting anticholinergics compared to short-acting beta-2 agonists

Please see the comparison listed above under the section 'Short-acting beta agonists'.

Short-acting anticholinergics compared to long-acting beta-2 agonists

No trials reported this comparison.

LONG-ACTING ANTICHOLINERGICS

No trials reported on this intervention group.

DISCUSSION

The primary aim of this review was to evaluate the effectiveness of inhaled bronchodilators in children and adults with cystic fibrosis (CF). We identified 18 trials with a total of 369 participants which

satisfied our criteria for eligibility. Inhaled bronchodilators have a variable response in people with CF. Some inhaled bronchodilators may be useful especially in individuals who have been shown to have bronchial hyperreactivity and bronchodilator responsiveness.

The design and quality of the trials varied greatly. These variations inevitably contributed to the inconsistencies in the conclusions of the different trials. Investigators used different dosages, dosage frequencies and inhaler devices to deliver the different bronchodilators they were evaluating. Only five trials were long-term and hence clinically relevant to a prescribing physician or to the individual prescribed them (Eggleston 1991; Hordvik 2002; Konig 1998; Turchetta 1996; Turchetta 1998). In six trials only a single dose of the agent being studied was given (Avital 1992; Dodd 2005; Sanchez 1992; Serisier 2007; Wiebicke 1990; Ziebach 2001).

Three trials selected participants with objective evidence of bronchodilator responsiveness (Sanchez 1992; Sanchez 1993; Salvatore 2002) and a further three trials used tests of bronchial hyperreactivity (bronchial hyper-responsiveness) with methacholine or histamine (Avital 1992; Eggleston 1991; Serisier 2007). One further trial excluded participants with a history of asthma (Konig 1998).

As one would expect, participants who showed either bronchial hyperreactivity or bronchodilator responsiveness, or both, were more likely to benefit from inhaled bronchodilators. In many trials the presence of bronchial hyperreactivity or bronchodilator responsiveness was not clarified and this may explain the different results observed in the trials. Bronchial hyperreactivity (excessive responsiveness of the airways to bronchoconstrictive stimuli) occurs in 24% to 51% of people with CF (Balfour-Lynn 2002). However, this does not reliably predict bronchodilator responsiveness which occurred in one trial in 25% of those with bronchial hyperreactivity (Eggleston 1991). Furthermore, neither bronchodilator responsiveness nor bronchial hyper-responsiveness is a reliable indicator of CF asthma (Balfour-Lynn 2002). People with CF are known to show both diurnal variation but also variability of airway responsiveness over time and the mechanisms for this variation are not fully understood (Holzer 1981).

In many trials, little or no information was provided about concomitant therapies, such as inhaled or oral corticosteroids, nebulised antibiotics or dornase alfa, or physiotherapy; yet all of these can alter lung function. With physiotherapy, the repeated forced expiratory manoeuvres (huffing) coupled with the proximal mobilisation of mucus by bronchodilatation may temporarily obstruct airway flow and so lung function tests measured immediately after bronchodilatation with chest physiotherapy may give falsely low readings (Hordvik 1996). Two out of the eighteen trials used additional chest physiotherapy as an adjunct during the trial (Hordvik 1996; Hordvik 1999).

Information about the severity of the lung disease in the participants was often absent. Some authors selected only participants

who had a baseline lung function within a predefined range. This limits the generalisability of the results to the whole CF population. Only three trials included a power calculation to estimate the minimum number of participants required to detect a significant response (Hordvik 2002; Konig 1998; Serisier 2007). The absence of evidence of a bronchodilator effect in many of the small trials may be a reflection of inadequate sample size (type II error).

Despite these problems, some conclusions can be drawn:

Firstly salbutamol (or albuterol) may be more beneficial than anticholinergics. Salbutamol is the most widely-used bronchodilator in people with CF and in the short-term, three out of six trials showed a significant improvement in FEV₁ after a single dose of salbutamol (Dodd 2005; Serisier 2007; Ziebach 2001); whereas three out of four trials comparing the anti-cholinergic ipratropium bromide to placebo showed no effect (Avital 1992; Pitcher 1982; Wiebicke 1990). Interestingly, both the trials using multiple doses of salbutamol showed an improvement in FEV₁ (Hordvik 1996; Hordvik 1999). However, there was no evidence that inhaled salbutamol improved FVC, FEF_{25–75} or peak flow in children or adults with CF or that they had any effect on symptoms. In the long term, one trial using salbutamol showed an improvement in peak expiratory flow in participants who had evidence of bronchial hyperreactivity at enrolment (Eggleston 1991). However, this trial did not detect any effect on FEV₁ or other spirometric indices or on symptom scores. The clinical significance of these results is therefore debatable.

Secondly salmeterol may be more beneficial than salbutamol. In theory, the longer duration of action of salmeterol should be advantageous in CF. The trials of salmeterol that compared it to either a placebo (Hordvik 1999; Salvatore 2002; Turchetta 1998) or salbutamol (Hordvik 1999; Hordvik 2002; Turchetta 1996) showed a significant improvement in FEV₁ and FEF_{25–75} the morning after a dose was given. One trial also showed a small improvement in nocturnal hypoxaemia with salmeterol compared to placebo (Salvatore 2002). The long-term trial of Hordvik showed significant differences in FEV₁ between participants receiving salmeterol compared to salbutamol after 12 and 24 weeks, but there was no effect on other spirometric measures (Hordvik 2002). They also reported reduced symptoms in those participants receiving salmeterol in the last 12 weeks of their long-term treatment but not in other periods. Laboratory trials of salmeterol have shown that it has other actions beyond relaxation of the smooth muscle of the airways. In vitro trials have shown that it reduced the damage to the respiratory mucosa caused by *Pseudomonas aeruginosa* (Dowling 1999) and non-typable *Haemophilus influenzae* (Dowling 1998).

Thirdly, ipratropium appeared to protect against methacholine induced bronchoconstriction in children with CF and proven bronchial hyperreactivity, but again the clinical relevance of this observation is unclear (Avital 1992). There were no long-term trials of the regular use of ipratropium in CF.

We found no published trials of fenoterol, formoterol or tiotropium in CF and the use of these agents cannot be supported.

In many of the trials we included, the primary measure of effect was the change in FEV₁ and other spirometric indices immediately after inhaling a bronchodilator. It is difficult to extrapolate from such short-term changes to long-term effects or to relate them to symptomatic improvement. Many of the changes in lung function, although statistically significant, were modest at approximately 5% change in FEV₁ and well within the 10% variation in lung function measurements which can occur spontaneously in CF. However, the observed changes were of a similar magnitude to those observed with other therapies used in CF, such as nebulised dornase alfa and nebulised tobramycin. The papers did not attempt to look at the effectiveness of bronchodilators in their hypothesised other roles, such as an aid to physiotherapy and sputum clearance, or as a tool to prevent symptoms during nebulised therapies such as tobramycin.

The long-term trials commented on the effect of bronchodilators on symptoms and quality of life. Symptom scoring is difficult, but although the trials did not show an improvement in symptoms, they did not show adverse effects. In general, the bronchodilators were safe and well-tolerated. However, a primary concern of people with CF is whether bronchodilators relieve symptoms and improve quality of life rather than just improving spirometry. There is a poor correlation between changes in lung function tests and symptoms in CF. Symptoms, although paramount to the person's quality of life, are an insensitive indicator of lung function (Konig 1982).

One of our secondary aims was to establish the most effective dosage regimen, but because of the heterogeneity of doses and dose frequencies this was not possible. It was possible though, to look at the effect of route of administration of the drug, for the eight trials in which a short-acting beta-2 agonist was compared to a placebo in the short term (Avital 1992; Dodd 2005; Hordvik 1996; Hordvik 1999; Pitcher 1982; Serisier 2007; Sheehan 2003; Ziebach 2001). All three trials using an MDI to deliver the drug or placebo found an increase in FEV₁ (Hordvik 1999; Serisier 2007; Sheehan 2003); whereas two of the four trials using a nebuliser found a significant increase in FEV₁ (Hordvik 1996; Ziebach 2001). In one trial the route of delivery was described as inhaled but no further detail was given (Dodd 2005). This suggests that the inhaled route via a MDI produced a more effective mode of administration, but the numbers involved are small and the participant's age is likely to have been important.

AUTHORS' CONCLUSIONS

Implications for practice

Despite the widespread, and often long-term, use of salbutamol (albuterol) and other bronchodilators in children and adults with CF, there is relatively little evidence to support this practice. Any benefit is most likely to be seen in those individuals who have objective evidence of bronchodilator responsiveness or bronchial hyperreactivity or both. We would suggest that before embarking on what may be lifelong bronchodilator therapy, an initial assessment should be carried out. This may be a test of bronchodilator responsiveness by measuring lung function before and after a test dose of salbutamol in a standardised way, looking for an increase in baseline FEV₁ of at least 10%, or it may be reasonable to assess bronchial hyperreactivity before considering bronchodilator therapy by measuring lung function before and after inhaled methacholine or histamine in a standardised way and looking for a fall in baseline FEV₁ of at least 10%. Similarly monitoring patients for diurnal variation or other evidence of atopy to define those with CF asthma may be useful. This approach would not only help identify those individuals who may benefit from long-term bronchodilator therapy, but also the important minority in whom bronchodilators may lead to a deterioration of lung function (Zach 1985).

All therapies should then have an initial trial period of several weeks where the benefit to the patient is assessed, not only by pulmonary function test but also by assessing the effect on quality of life, exercise capacity, rates of respiratory symptoms etc. After this trial period, if the therapy is not beneficial or detrimental, then it should be stopped. Similarly as bronchodilator responsiveness varies with time in people with CF, if therapy is continued for many years such assessments should be repeated periodically (Holzer 1981). If a bronchodilator is to be used, there is more evidence to support

the use of beta-2 agonists than anticholinergic agents. In those people with CF with evidence of bronchodilator responsiveness or a history suggestive of CF asthma, or both, a trial of the long-acting beta-2 agonist salmeterol would be appropriate.

Implications for research

Further research is required to clarify the benefits and the clinical role of bronchodilators in people with CF. On the basis of the available evidence, we would suggest long-term trials of the short-acting beta-2 agonist salbutamol (albuterol) and the long-acting beta-2 agonist salmeterol. Such trials should be adequately powered to detect a clinically, as well as a statistically, significant difference in FEV₁ as well as consider the important secondary outcomes of effect on exercise capability and QOL scores. They should ideally be of a double-blinded placebo-controlled design. The design and analysis of such trials should include factors such as disease severity, concomitant therapies, and evidence of bronchial responsiveness and hyperreactivity. Short-acting beta-2 agonists are widely used as a prelude or adjunct to physiotherapy in the belief that they improve sputum clearance. We suggest specific trials are required to provide objective evidence of such effect.

Due to the widespread use of these therapies such a RCT may not be possible or ethical but alternative research methods may include withdrawal trials or trials which compare 'responders' to non-responders' or CF asthmatics to non-asthmatics.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Avital 1992

Methods	Placebo controlled cross-over trial over 3 days	
Participants	11 participants (5 males) with history of RAD, mean (SD) age 10.5 (2.3) years SK score 60 - 95	
Interventions	Baseline spirometry Treatment with single dose of either nebulised placebo, 2.5 mg salbutamol or 0.25 mg ipratropium 45 minutes later FEV ₁ Methacholine challenge doubling concentrations of inhaled methacholine 5 minutes apart with FEV ₁ taken after each increase	
Outcomes	Drop of greater than 20% of FEV ₁ from baseline to give the concentration of methacholine or maximum concentration of methacholine	
Notes		
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Low risk	Stated as double-blind.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Dodd 2005

Methods	Placebo-controlled cross-over trial over 2 days
Participants	9 adults, no details given of age or sex of participants
Interventions	Maximal cycle ergometry on 2 days with either single dose of inhaled salbutamol (600 mcg) compared to placebo taken before exercise Doses not given
Outcomes	Change in FEV ₁ , exercise duration in seconds and maximal oxygen uptake in litres/min. (Exercise capacity data not included in analysis)

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Described as randomised but no details given of method.
Allocation concealment (selection bias)	Low risk	Prepared by external attendant.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No details of method given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	More than 15 % of participants lost to follow up.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Duncan 1982

Methods	Placebo-controlled cross-over trial over 3 days
Participants	16 participants, no details given of age or sex of participants
Interventions	Nebulised salbutamol 0.5 mg or saline prior to physiotherapy
Outcomes	Spirometry (FEV ₁ , FVC and PEFR) at baseline, 5 minutes, 1 hour, 2 hours and 3 hours after physiotherapy Weight of sputum produced in three hours after physiotherapy

Notes

Risk of bias

Duncan 1982 (Continued)

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Described as randomised but no details given of method.
Allocation concealment (selection bias)	Unclear risk	No methods described.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No methods given, but stated single-blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Eggleston 1991

Methods	Placebo-controlled cross-over trial over 2 months
Participants	27 participants (13 males), data from 22 participants only, age range 6 - 25 years SK scores 62 - 90 14/16 responders, 8/11 non-responders completed trial
Interventions	Subdivided into responders or non-responders by methacholine challenge. Randomised to either albuterol 90 microgram qds compared to placebo by MDI for 4 weeks
Outcomes	Changes in spirometry (FEV ₁ , FVC, PEFR, FEF ₂₅₋₇₅) measured at baseline, 2 and 4 weeks Daily PEFR measurements Symptom diaries
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.

Eggleston 1991 (Continued)

Blinding (performance bias and detection bias) All outcomes	Low risk	Methods of double-blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	More than 15% lost to follow up, but reasons given.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Hordvik 1996

Methods	Placebo-controlled cross-over trial over 2 days
Participants	24 participants (14 males), age range 13 - 33 years NIH score 49 - 89
Interventions	Nebulised albuterol 0.5% (5 mg) or placebo to the first 3 of 4 CPT sessions at 0700 hours, 1100 hours, 1500 hours and 1900 hours
Outcomes	Changes in spirometry (FVC FEV ₁ FEF ₂₅₋₇₅) measured pre- and 45 minutes post 0700 hours and 1500 hours, pre- 1900 hours and pre- 0700 hours next morning
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Low risk	Double-blind, methods given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up, less than 15% lost, reasons given.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Hordvik 1999

Methods	3-way cross-over trial over 3 days
Participants	20 participants (7 males), age range 13 - 36 years NIH score 43 - 85
Interventions	Day 1: 2 puffs of albuterol MDI (180 mcg prior to 4 sessions of CPT) Day 2: 2 puffs of salmeterol (42 mcg) before first and last sessions of CPT and placebo before sessions 2 and 3 Day 3: 2 puffs of placebo. CPT sessions at 0700 hours, 1100 hours, 1500 hours and 2100 hours
Outcomes	Changes in spirometry (FEV ₁ , FVC, FEF ₂₅₋₇₅) measured pre- and 45 minutes post 0700 hours and 1500 hours, pre- 1900 hours and pre- 0700 hours next morning
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Low risk	Double-blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up, less than 15% lost, reasons given.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Hordvik 2002

Methods	Placebo and dual intervention cross-over trial, short-term arm 4 weeks; long-term arm 26 weeks
Participants	44 participants (16 males), age range 13 - 37 years 36 participants finished short-term arm Initial FEV ₁ 69%
Interventions	Received either nebulised albuterol 2.5 mg bd and placebo discus or salmeterol 100 mcg bd by discus and placebo nebuliser

Hordvik 2002 (Continued)

Outcomes	Changes in spirometry (FEV ₁ , FVC, FEF _{25–75})measured at baseline, 4 weeks, 12 weeks and 24 weeks Side effects Need for antibiotic interventions	
Notes		
<i>Risk of bias</i>		
Bias	Authors’ judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code
Allocation concealment (selection bias)	Low risk	External administrator
Blinding (performance bias and detection bias) All outcomes	Low risk	Methods given
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up, more than 15% lost, reasons given
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Konig 1998

Methods	Placebo-controlled cross-over trial over 12 months	
Participants	30 participants (12 males), age range 5 - 17 years 21 participants finished trial Anyone with a history of asthma was excluded Mean SK score 67.9	
Interventions	Albuterol 180 mcg via MDI versus placebo bd for 6 months	
Outcomes	Changes in spirometry (PEFR,FEV ₁ , FVC, FEF _{25–75} , FEF ₇₅) measured at start and end of 6 months Side effects Number of hospitalizations	
Notes		
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement

König 1998 (Continued)

Random sequence generation (selection bias)	Low risk	Table of random numbers.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Low risk	Double-blind, methods given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up, more than 15% lost, reasons given.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Pitcher 1982

Methods	Dual intervention cross-over trial over 2 - 6 months
Participants	30 children (no details of number of males and females), age range 7 - 16 years
Interventions	Single dose of nebulised saline, salbutamol (5 - 10 mg) or ipratropium (250 - 500 mcg)
Outcomes	Changes in spirometry (FEV ₁ , FVC, FEF ₂₅ , PEF _R) measured before and 15 minutes after treatment
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Described as randomised but no details given of method.
Allocation concealment (selection bias)	Unclear risk	No methods given.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No details given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.

Pitcher 1982 (Continued)

Other bias	Low risk	No other bias identified.
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Salvatore 2002

Methods	Placebo-controlled cross-over trial over 4 days
Participants	23 participants 14 responders (8 males), mean age 21.4 years, age range 18.5 - 36.6 years 9 non-responders (5 males), mean age 22.1, range 18.8 - 25.4
Interventions	Day 1: albuterol challenge (200 mcg via MDI) to determine responders Day 2: 45 minutes before bed 4 puffs of inhaled salmeterol (84 mcg) or placebo. Overnight transcutaneous oxygen saturations measured Day 3: morning spirometry Day 4: repeat of Day 2
Outcomes	Changes in spirometry (FEV ₁ , FEF ₂₅₋₇₅) Changes in oxygen saturation (data not included in analysis)
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Described as randomised but no details given of method
Allocation concealment (selection bias)	Unclear risk	No method given.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No method given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Sanchez 1992

Methods	Dual intervention cross-over trial over 3 days
Participants	12 participants (6 males), aged 9 - 22 years SK score 45 - 90
Interventions	Received either single dose nebulised placebo, albuterol 5.0 mg or ipratropium 0.25 mg
Outcomes	Changes spirometry (FVC, FEV ₁ , FEF ₂₅₋₇₅ , FEF ₅₀ , PEFR, FRC, RV, R _{aw}) measured at baseline and 45 minutes post-treatment
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Low risk	Method given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Sanchez 1993

Methods	Dual intervention cross-over trial over 2 days
Participants	9 participants (4 males), age range 6 - 22 years Only those with previous positive response to salbutamol SK score 35 - 90
Interventions	Randomised to: Sequence A: salbutamol 200 mcg, salbutamol 400 mcg and ipratropium 80 mcg by MDI and aerochamber separated by 45 minutes; or Sequence B: ipratropium 80 mcg, salbutamol 200 mcg, salbutamol 400 mcg
Outcomes	Changes in spirometry (FVC, FEV ₁ , FEF ₂₅₋₇₅ , FEF ₅₀ , PEFR, FRC, RV, R _{aw}) measured at baseline, before each drug in the sequence and at 4 hours and 8 hours from baseline Only 4 hour data used

Sanchez 1993 (Continued)

Notes		
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Low risk	Methods given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Serisier 2007

Methods	Placebo-controlled cross-over trial over 2 days	
Participants	12 participants (6 males), age range 17 - 39 years Histamine challenge test was performed to assess bronchial hyper-reactivity but no information given	
Interventions	Randomised to either single dose salbutamol 600 mcg or placebo via MDI with spacer before cycle ergometry	
Outcomes	Changes in spirometry (FEV ₁ and maximal exercise) Exercise capacity (data not included in analysis)	
Notes		
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.

Serisier 2007 (Continued)

Blinding (performance bias and detection bias) All outcomes	Low risk	Methods given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up, less than 15 % lost to follow up, reasons given.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

Sheehan 2003

Methods	Placebo-controlled parallel trial
Participants	Albuterol group: 48 infants with CF aged 9 to 250 weeks (88 studies recorded) Placebo group: 15 infants with CF aged 17 to 254 weeks (15 studies recorded)
Interventions	Albuterol in a dose of 2 puffs (180 mcg) MDI and spacer followed by lung inflation to 25 cm H ₂ O given at 2 minute intervals until heart rate increased by 10% (maximum of 8 puffs) compared to placebo (propellant only) in dose of 2 puffs at 2 minute intervals (maximum 6 puffs)
Outcomes	Maximal expiratory flow (FEF ₂₅₋₇₅ , FEF ₇₅ , FEF ₅₀ , FVC, FEV _{0.5})
Notes	Albuterol administered using metered dose inhaler with spacer (Aerochamber MV)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not discussed.
Allocation concealment (selection bias)	Unclear risk	Not discussed.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	Compared to placebo in same dose, but maximum number of puffs from albuterol and placebo different, so participants would be able to tell which group they were in if needed maximum number of puffs. Clinicians would need to know whether participants were receiving albuterol or placebo to regulate maximum number of puffs allowed
Incomplete outcome data (attrition bias) All outcomes	Low risk	Withdrawals and dropouts not discussed but data presented for all recruited

Sheehan 2003 (Continued)

Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Unclear risk	Published as abstract only

Turchetta 1996

Methods	2 interventions randomised controlled cross over trial
Participants	22 children (9 male, mean (SD) age 12 (4) years)
Interventions	Baseline lung function and then either Salmeterol 2 puffs bd or Salbutamol 2 puffs tds for 30 days and then 30 days of the other treatment. No washout period
Outcomes	Lung function tests at baseline and after 60 days (FEV ₁ , FVC, FEF ₂₅₋₇₅)
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No details of methods given.
Allocation concealment (selection bias)	Unclear risk	No details of methods given.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No details of methods given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All completed study.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Unclear risk	Published only as abstract

Turchetta 1998

Methods	Randomised placebo-controlled parallel trial
Participants	16 children (11 males, mean (SD) age 13.9 (3.1) years, mean (SD) FEV ₁ 72% (19%); in good clinical condition with no intercurrent infection on dietary supplements, pancreatic enzymes, mucolytics and aerosol antibiotics
Interventions	Salmeterol 2 puffs bd via MDI and spacer or placebo for one month

Turchetta 1998 (Continued)

Outcomes	Change in FEV ₁ from baseline	
Notes		
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No methods given.
Allocation concealment (selection bias)	Unclear risk	No methods given.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No methods given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All complete.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Unclear risk	Published only as abstract.

Wiebicke 1990

Methods	Randomised, single-blind, placebo-controlled cross over	
Participants	11 participants (8 males), age range 8 - 29 years, mean (SEM) age 18.1 (2.0) years. Normal nutritional status. Schwachman score ranged from 53 to 97. Other medication discontinued before study commenced	
Interventions	2 ml placebo (0.9 % hypertonic saline solution) or 2 ml ipratropium bromide solution (Atrovent® = 250 µg in 1 ml hypertonic saline solution) on different days in the space of 2 weeks	
Outcomes	Lung function (FEV ₁ , FEF _{25–75%} , RV, TLC) measured before, then 30, 60 and 120 min after inhalation	
Notes		
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Stated as randomised, method not described.

Wiebicke 1990 (Continued)

Allocation concealment (selection bias)	Unclear risk	Not discussed.
Blinding (performance bias and detection bias) All outcomes	Low risk	Single-blind, does not state who was blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data for all 11 participants provided.
Selective reporting (reporting bias)	Unclear risk	Protocol not available to check against published paper, no selective reporting identified

Ziebach 2001

Methods	Dual intervention cross-over trial over 4 days
Participants	17 participants (6 males), age range 7 - 33 years, 16 participants analysed Baseline FEV ₁
Interventions	Randomised to single dose nebulised placebo, salbutamol 5 mg, ipratropium 0.25 mg or combination
Outcomes	Changes in spirometry (FVC, FEV ₁ , FEF ₂₅₋₇₅ , FEF ₅₀ , TGV, R _{aw}) measured at baseline and 30 minutes later
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Central code.
Allocation concealment (selection bias)	Low risk	External administrator.
Blinding (performance bias and detection bias) All outcomes	Unclear risk	No details given.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow up, less than 15% lost, but reasons given.
Selective reporting (reporting bias)	Low risk	No selective reporting identified.
Other bias	Low risk	No other bias identified.

bd: twice daily
 CPT: chest physiotherapy
 FEV₁: forced expiratory volume at one second
 MDI: metered dose inhaler
 NIH: National Institute of Health score
 PEF: peak expiratory flow rate
 qds: four times daily
 SK score: Shwachman-Kulczycki score
 RAD: reversible airway disease
 tds: three times daily

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Avital 1991	Not a predefined bronchodilator - oral route
Chua 1989	Not a predefined outcome - ticarcillin-induced bronchoconstriction
Chua 1993	Not a predefined outcome - ticarcillin-induced bronchoconstriction
Finnegan 1989	Not a predefined route of administration - intravenous
Gulliver 2003	Not a bronchodilator - inhaled antibiotic
Kattan 1980	Not a predefined bronchodilator
Kohler 2004	Not a predefined outcome
Konig 1995	Not an RCT
Konig 2005	Not a predefined outcome
Kovaleva 2000	Not a relevant intervention, combination of enzymotherapy and nebulizer therapy
Kraemer 1996	Study discussing relationship to physiotherapy
Larsen 1979	Not an predefined bronchodilator
Liedtke 1996	Not a predefined outcome
Maayan 1989	Not a predefined bronchodilator
Marks 1998	Study comparing delivery devices
Mortensen 1993	Not a predefined outcome - used mucociliary clearance

(Continued)

Nikolaizik 1996	Not a bronchodilator - inhaled antibiotic
Orlik 2002	Not a predefined outcome - looked at chest physiotherapy
Peckham 1995	Outcomes not relevant to this review
Robuschi 1990	Not a bronchodilator
Shapiro 1976	Not a predefined bronchodilator
Sivan 1990	Not a predefined bronchodilator
Tecklin 1976	Not a predefined bronchodilator
Van Asperen 1988	Not an RCT
Weintraub 1989	Not a predefined bronchodilator

RCT: randomised controlled trial

Characteristics of studies awaiting assessment *[ordered by study ID]*

Bessaci 2008

Methods	Described as a double-blind crossover study, randomisation process not clear
Participants	20 children (6 - 18 years old)
Interventions	2x 168 day treatment periods (<12 years old: SFP 50/100 µg bid, >12 years: 50/250 µg bid versus placebo). Separated by 6 week washout period
Outcomes	clinical parameters, lung function, quantification of inflammatory mediators in sputum (IL-8, MMP9, elastase, TIMP-1), nasal potential difference
Notes	PUBLISHED only as abstract

NCT00737100

Methods	Randomized double-blind safety/efficacy study (Phase 2). Duration: 12 weeks. Parallel design. Multi-centre (107 study locations). Study Start Date: September 2008 Primary Completion Date: April 2010 (final data collection date for primary outcome measure)
Participants	Enrollment: 510 participants
Interventions	3-arm study: Group 1: Tiotropium Respimat 2.5 mcg; patient to receive low dose tiotropium once daily Group 2: Tiotropium Respimat 5 mcg; patient to receive high dose tiotropium once daily Group 3: Placebo Respimat: patient to receive placebo once daily
Outcomes	Data to be reported at end of 12 weeks Primary Outcome Measures Change from baseline in FEV ₁ AUC Change from baseline in trough FEV ₁ Secondary Outcome Measures Change from baseline in static lung hyperinflation as measured by RV/TLC Change from baseline in CFQ scores Change from baseline in FVC AUC Change from baseline in trough FVC Change from baseline in pre-bronchodilator FEF _{25–75} % predicted Proportion of patients with at least 1 pulmonary exacerbation
Notes	Information provided by sponsors Boehringer Ingelheim Pharmaceuticals. ClinicalTrials.gov identifier: NCT00737100 Official title: A Randomized, Double-blind, Placebo-controlled Parallel Group Study to Investigate the Safety and Efficacy of Two Doses of Tiotropium Bromide (2.5 Mcg and 5 Mcg) Administered Once Daily Via the Respimat Device for 12 Weeks in Patients With Cystic Fibrosis

AUC: area under the curve

bid: twice daily

CFQ: cystic fibrosis questionnaire

FEF_{25–75}: forced expiratory flow from 25% to 75% of vital capacity

FEV₁: forced expiratory volume at one second

FVC: forced vital capacity

RV: residual volume

SFP: salmeterol-fluticasone propionate

TLC: total lung capacity

Characteristics of ongoing studies [ordered by study ID]

NCT01179347

Trial name or title	NCT01179347: A Randomised, Double-blind, Placebo-controlled Parallel-group Trial to Confirm the Efficacy After 12 Weeks and the Safety of Tiotropium 5 Mcg Administered Once Daily Via the Respimat® Device in Patients With Cystic Fibrosis
Methods	Randomized double-blind placebo-controlled safety/efficacy study (Phase 3). Parallel design. Duration: 12 weeks. Multi-centre (55 study locations).
Participants	Estimated enrolment: 389
Interventions	Group 1: 2.5 mcg tiotropium: 2 inhalations once daily delivered with Respimat® inhaler Group 2: placebo: 2 inhalations once daily delivered with Respimat® inhaler
Outcomes	Data to be collected at baseline and 12 weeks Primary Outcome Measures Change from baseline in percent predicted FEV ₁ AUC0-4H Change from baseline in percent predicted trough FEV ₁ Secondary Outcome Measures Change from baseline in % predicted trough FVC (AUC0-4H) at 12 weeks. Change from baseline in % predicted trough FVC at 12 weeks Change from baseline in pre-bronchodilator mean FEF ₂₅₋₇₅ Change from baseline in CFQ-R Proportion of patients with at least 1 pulmonary exacerbation
Starting date	September 2010
Contact information	Boehringer Ingelheim Call Center: 1-800-243-0127 or clintrriage.rdg@boehringer-ingelheim.com
Notes	Estimated primary completion date: December 2011 (final data collection date for primary outcome measure)

AUC: area under the curve

CFQ-R: Cystic Fibrosis Questionnaire revised

FEF₂₅₋₇₅: forced expiratory flow from 25% to 75% of vital capacity

FEV₁: forced expiratory volume at one second

FVC: forced vital capacity

DATA AND ANALYSES

This review has no analyses.

ADDITIONAL TABLES

Table 1. Comparison of interventions

Study	Drug A	Drug B	Control	Dose	Frequency	Type of inhaler	Notes
Avital 1992	Salbutamol	Ipratropium	Placebo	Salbutamol: 2.5 mg Ipratropium: 0.25 mg	Single dose		Cross-over trial. Methacholine challenge doubling concentrations of inhaled methacholine 5 minutes apart with FEV ₁
Dodd 2005	Salbutamol		Placebo	Salbutamol: 600 mcg Ipratropium: 0.25 mg	Single dose		Cross-over trial.
Duncan 1982	Salbutamol		Saline	Salbutamol: 0.5 mg	prior to physiotherapy (3 days)		Cross-over trial.
Eggleston 1991	Albuterol		Placebo	Albuterol: 90 mcg	4x daily (4 weeks)	MDI	Cross-over trial. Participants subdivided into responders or non-responders by methacholine challenge
Hordvik 1996	Albuterol		Placebo	5mg	3x prior to CPT (1 day)		Cross-over trial.
Hordvik 1999	Albuterol	Salmeterol	Placebo	Albuterol: 180 mcg Salmeterol: 42 mcg	Albuterol: 4x prior to CPT (1 day) Salmeterol: 2x prior to 1st and 4th	MDI	3-way cross-over trial

Table 1. Comparison of interventions (Continued)

					CPT (1 day) Placebo: 4x prior to CPT (1 day)		
Hordvik 2002	Albuterol & placebo	Salmeterol & placebo		Albuterol: 2.5 mg Sal-meterol: 100 mcg	both given 2x daily (Short-term arm 4 weeks; long-term arm 26 weeks)		Cross-over trial. Abuterol nebulised and placebo discus; Sal-meterol discus and placebo nebulised
Konig 1998	Albuterol		Placebo	Albuterol: 180 mcg	2x daily (6 months)	MDI	Cross-over trial.
Pitcher 1982	Salbutamol	Ipratropium	Saline	Salbu-tamol: 5 - 10 mg Iprat-ropium: 250 - 500 mcg	Single dose (2 - 6 months)		Cross-over trial.
Salvatore 2002	Salmeterol		Placebo	Salmeterol: 84 mcg	1x daily (45 mins before bed), repeat 48 hours later		Cross-over trial. Day 1: albuterol challenge (200 mcg via MDI) to determine responders
Sanchez 1992	Albuterol	Ipratropium	Placebo	Albuterol: 5 mg Ipratropium: 2.5 mg	Single dose of each (3 days)		Cross-over trial.
Sanchez 1993	Salbutamol	Ipratropium		Salbutamol: 200 - 400 mcg Ipratropium: 80 mcg	Single dose of each sequence, dose of each drug separated by 45 mins (2 days)	MDI & ae-rochamber	Cross-over trial. Sequence A: salbutamol 200 mcg, salbutamol 400 mcg and ipratropium 80 mcg; or Sequence B: ipratropium 80 mcg, salbu-

Table 1. Comparison of interventions (Continued)

							tamol 200 mcg, salbutamol 400 mcg
Serisier 2007	Salbutamol		Placebo	Salbutamol: 600 mcg	Single dose prior to cycle ergometry.	MDI with spacer	Cross-over trial. Histamine challenge test was performed to assess bronchial hyper-reactivity but no information given
Sheehan 2003	Albuterol		Placebo	Albuterol: 180 mcg	Albuterol: 2 minute intervals until heart rate increased by 10% (maximum of 8 puffs); Placebo: 2 puffs at 2 minute intervals (maximum 6 puffs).	MDI with spacer (Aerochamber MV)	Parallel trial.
Turchetta 1996	Salbutamol	Salmeterol		Salbutamol: 2 puffs Salmeterol: 2 puffs	Salbutamol: 3x daily (30 days) Salmeterol: 2x daily (30 days)		Cross-over trial. No washout period.
Turchetta 1998	Salmeterol		Placebo	2 puffs	2x daily (1 month)	MDI with spacer	Parallel trial.
Wiebicke 1990	Ipratropium bromide solution		0.9 % hypertonic saline solution	Ipratropium bromide: 2 ml Hypertonic saline: 2 ml	Single dose of each (over 2 weeks)		Cross-over trial. Ipratropium bromide solution (Atrovent® = 250 µg in 1 ml hypertonic saline solution)

Table 1. Comparison of interventions (Continued)

Ziebach 2001	Salbutamol	Ipratropium	Placebo	Salbutamol: 5 mg Ipratropium: 0.25 mg	Single dose of each (4 days)		Cross-over trial. Randomised to nebulised placebo, salbutamol, ipratropium or combination
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WHAT'S NEW

Last assessed as up-to-date: 4 April 2011.

Date	Event	Description
5 April 2011	New search has been performed	A search of the Group's Cystic Fibrosis Trials Register identified five references to four studies potentially eligible for inclusion in the review. Two studies were excluded (Kovaleva 2000 ; Tecklin 1976), one phase 2 study has been completed but not yet published and is listed as 'Awaiting Assessment' (NCT00737100) and the final study is still ongoing (NCT01179347).

HISTORY

Protocol first published: Issue 1, 2002

Review first published: Issue 4, 2005

Date	Event	Description
19 October 2009	Amended	A table comparing all interventions included in this review has been added to the additional tables for clarification
19 October 2009	New search has been performed	A search of the Cystic Fibrosis Trials Register identified two new references to one study which has been listed as 'Awaiting classification' until more information about the study design is available (Bessaci 2008).
12 June 2008	Amended	Converted to new review format. Comments from the Group's Medical Statistician have been addressed A new plain language summary has been drafted in line with The Cochrane Collaboration's latest guidance

(Continued)

12 June 2008	New search has been performed	A search of the Cystic Fibrosis & Genetic Disorders Group's Trials Register identified 15 new references, 14 were eligible for the review. Of these 14 references, six were included in the review (Dodd 2005 ; Serisier 2007 ; Sheehan 2003 ; Turchetta 1996 ; Turchetta 1998 ; Wiebicke 1990). Two of these were full articles of previously included abstracts (Dodd 2005 ; Serisier 2007) The remaining eight references were excluded (Avital 1991 ; Gulliver 2003 ; Kohler 2004 ; Konig 2005 ; Kraemer 1996 ; Liedtke 1996 ; Nikolaizik 1996 ; Peckham 1995).
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CONTRIBUTIONS OF AUTHORS

Dr Halfhide co-ordinated the changing of the protocol into a full review. This included undertaking searches, appraising quality of papers, data extraction, entering data into RevMan, analysis, interpretation of data and writing the review. Dr Halfhide acts as guarantor of the review.

Dr Couriel originally conceived the idea for the protocol and review. He was the lead author of the protocol. He provided advice throughout the selection of the eligible studies, data interpretation and the writing of the review.

Dr Evans provided useful comments and ideas throughout conception of the protocol. During the review she provided help with appraising the quality of the studies, data extraction and interpretation of results.

DECLARATIONS OF INTEREST

None known.

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

There was a post hoc change to the types of bronchodilators to be included in the review. There is now a fourth category 'long-acting anticholinergics agents (tiotropium)' due to the fact that these preparations are now commercially available.

INDEX TERMS

Medical Subject Headings (MeSH)

Adrenergic beta-Agonists [*therapeutic use]; Albuterol [analogs & derivatives; therapeutic use]; Bronchial Hyperreactivity [*drug therapy; etiology]; Bronchodilator Agents [*therapeutic use]; Cystic Fibrosis [*complications]; Ipratropium [therapeutic use]; Randomized Controlled Trials as Topic

MeSH check words

Adult; Child; Humans